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Oral Abstract Session

Erdafitinib in patients with FGFR-altered tumors: Results from the NCI-COG Pediatric MATCH trial arm B (APEC1621B).

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Background: The NCI-Children's Oncology Group (COG) Pediatric Molecular Analysis for Therapy Choice (MATCH) trial assigns patients aged 1-21 years with relapsed or refractory solid tumors, CNS tumors, lymphomas, and histiocytic disorders to phase 2 treatment arms of molecularly-targeted therapies based on genetic alterations in their tumor. Arm B (APEC1621B) evaluated the orally bioavailable FGFR inhibitor erdafitinib in patients whose tumors harbored activating alterations in FGFR 1/2/3/4.

Methods: Patients received oral erdafitinib 4.7 mg/m²/dose (Max dose: 8 mg) daily for 28-day cycles until disease progression or drug related dose limiting toxicity (Max 26 cycles). Disease assessments were performed every 2-3 cycles. The primary endpoint was objective response rate; secondary endpoints included progression free survival (PFS). **Results:** A total of 20 evaluable patients (median age: 15 yrs) were enrolled from June 2018 to July 2022 in this first study of erdafitinib in pediatric patients. The most common diagnosis (n = 17) was glioma or glioneuronal tumor (11 low grade and 6 high grade). Three patients had non-CNS solid tumors (1 each with neuroblastoma, osteosarcoma, and rhabdomyosarcoma). FGFR alterations detected consisted of activating hotspot mutations in *FGFR1* (n = 16), *FGFR2* (n = 1), *FGFR4* (n = 1), and *FGFR1* fusions (n = 2; *FGFR1:TACC1*, *FGFR1:TACC3*). Partial responses were confirmed by central review in 2/20 patients (10%, 90%CI: 3.4%, 26.2%), both with low grade gliomas or glioneuronal tumors (LGG, LGGNT) and *FGFR1* hotspot mutations (K687E and N577K). A best response of stable disease (SD) was observed in 6 additional patients with gliomas with a median duration of SD of 6.5 cycles. Six-month PFS was 45% (95% CI: 23.1%, 64.7%) and six-month OS was 89.7% (95%CI: 64.8%, 97.3%). Treatment related adverse events included those previously reported with FGFR inhibitors including hyperphosphatemia and nail changes or infections. Grade 1 vision changes were also reported. There was one Grade 3 spinal cord compression and one Grade 4 intracranial hemorrhage. **Conclusions:** Erdafitinib was generally well tolerated in this heavily pre-treated cohort of children with relapsed tumors harboring FGFR alterations. Partial responses or stable disease were observed in the majority (6/11, 54%) of patients with *FGFR1*-mutant low grade gliomas or glioneuronal tumors, suggesting the possible benefit of FGFR inhibitor therapy for these patients. Further studies would be necessary to more thoroughly evaluate the activity of erdafitinib for patients with other histologies and FGFR alteration types or combination approaches that include other targeted and/or cytotoxic agents. Clinical trial information: NCT03210714. Research Sponsor: U.S. National Institutes of Health.